

*“In theory, there is no difference
between
theory and practice. In practice,
there is.”*

— Yogi Berra (attributed)

Abstract

Multiagent AI systems—autonomous coding assistants, research pipelines, financial decision engines—have moved from prototype to production in under two years. With them comes a new class of security concern: attacks that target not data or infrastructure but the *reasoning processes* of AI agents. Prompt injections that propagate through delegation chains, trust relationships that launder adversarial influence, and coordination mechanisms vulnerable to strategic manipulation all represent cognitive attack surfaces absent from traditional security models.

The **Cognitive Integrity Framework (CIF)** is developed across a three-part series: formal treatment (Part 1), running code and experiments (Part 2), and the present unified paper (Part 3+4) combining practitioner guidance with cross-domain application. **Part 1** establishes mathematical foundations—a trust calculus with provably bounded delegation, defense composition algebras with multiplicative detection guarantees, and information-theoretic limits on attack stealth. **Part 2** provides computational validation: eight implemented defense modules, 3,535 tests, a 950-attack corpus spanning four threat categories, parametric architecture-aware simulation across four production multiagent topologies, and a category-theoretic formalization of defense composition (Theorems CT.1–CT.3). The cross-domain applications section (§9–§10, originally “Part 4”) applies the framework via the integrated CIF-AD-OODA model across ten

critical operational domains.

This paper (Part 3+4, unified) is simultaneously a qualitative practitioner guide and a cross-domain application study. The practitioner section (§1–§8) synthesizes Parts 1 and 2 into accessible language, situates the formal results against current deployment practice, and gives practical recommendations for teams that build and run multiagent AI systems. The applications section (§9–§10) applies the framework across ten critical domains. No formal prerequisites are assumed; for proofs and definitions see Part 1, for empirical results see Part 2.

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This is Part 3+4 of the three-part *Cognitive Security for Multiagent Operators* series:

- ▶ **Part 1** (DOI: 10.5281/zenodo.22134544): Formal foundations and theoretical analysis
- ▶ **Part 2** (DOI: 10.5281/zenodo.22134546): Computational validation and implementation
- ▶ **Part 3+4** (this paper): Practitioner guidance (§1–§8) and cross-domain CIF-AD-ODA applications (§9–§10)

All source code, tests, and analysis scripts are maintained at
https://github.com/docxology/cognitive_integrity.